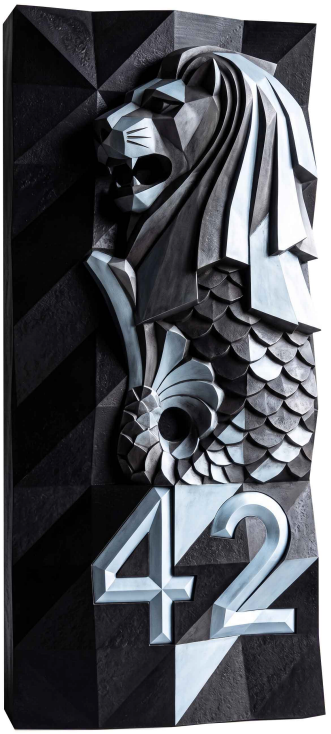


Discovery Piscine Side Project - 4

The Enigma Word Scramble

A 42 Singapore SUTD initiative

Summary: Build a mini word game

Version: *0*

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Chapter I

Preamble

Alan Mathison Turing (1912-1954) was a British mathematician, logician, and cryptanalyst who is widely regarded as the father of theoretical computer science and artificial intelligence.

In 1936, Turing published a groundbreaking paper introducing the concept of the "Turing machine," an abstract computing device capable of performing any computable algorithm. This theoretical model laid the fundamental blueprint for modern computers, proving that a single "Universal Turing Machine" could simulate any other machine by reading and writing symbols on an infinitely long tape.

During World War II, Turing's intellect became a vital asset to the Allied forces. Working at Bletchley Park, Britain's codebreaking center, he led the cryptanalysis section responsible for deciphering German naval communications. He designed the "Bombe," an electromechanical device that greatly accelerated the cracking of the notoriously complex Enigma ciphers. His work provided actionable intelligence that played a crucial role in defeating the Axis powers and significantly shortened the war.

After the war, Turing continued to push the boundaries of technology by proposing the first complete specification for an electronic stored-program all-purpose digital computer, known as the Automatic Computing Engine (ACE). He also turned his attention to philosophy and the mind, publishing a seminal 1950 paper that introduced the "Turing Test"-an experiment designed to set a standard for artificial intelligence by determining if a machine could successfully deceive a human into believing it was also human.

Despite his heroic contributions to his country and science, Turing's life ended in tragedy. In 1952, he was prosecuted by the British government for homosexual acts, which were illegal at the time. To avoid prison, he was forced to accept chemical castration via hormone treatment. He died from cyanide poisoning two years later at the age of 41, an event widely ruled as suicide.

Turing's legacy has only grown since his death. He was posthumously granted a royal pardon in 2013 by Queen Elizabeth II, and his visionary concepts remain the absolute foundation of modern computation, complexity theory, and the machines we use every single day.

Chapter II

General rules

- Read the subject completely. Do not rush. Read the instructions carefully from start to finish before writing a single line of code or creating your repository.
- Peer-to-peer learning is your first resource. If you have a question, ask your right neighbor. If you still have a question, ask your left neighbor. You are encouraged to search the internet and use AI.
- The Staff is not your search engine. You should only reach out to the pedagogical team after you have exhausted all other options and discussed the issue with your peers.
- Must use python!
- Have fun. This is your project. Make it yours!
- Vibe coding is allowed. Chill and, again, just have fun.

Chapter III

Specific instruction for today

- Upon completion, present your work to your cohort.

Chapter IV Exercise 00:
The Enigma Word Scramble

	Exercise 00
Exercise 00: The Enigma Word Scramble	
Turn-in directory : you need to figure this out/	
Files to turn in : `fenigma_scramble.py`	
Allowed: Vibe Coding	

Concept
A mini-game where the program selects a secret word, scrambles the letters, and forces the user to guess the original word

Learning:
Lists, Strings, and the `random` module.

Teamwork Mechanic:

1. Create a list of at least 10 vocabulary words related to ITE or technology.
2. Prompt the AI to figure out how to shuffle the letters of a string and display the scrambled version.

Specific Instructions & Checklist:
[ ] The script must import the `random` module.
[ ] The script must randomly select one word from Student A's list.
[ ] The script must scramble the letters of that word and print: "Guess the scrambled word: [scrambled_word]"
[ ] The script must use a `while` loop to let the user keep guessing until they type the correct word.
[ ] The AI Prompt Check: Tell the AI to ensure the program ignores uppercase/lowercase mistakes (e.g., typing "PYTHON" or "python" both count as correct).

Exciting, right?

Chapter V

Submission and peer-evaluation

Present your work to the world



`The evaluation process will happen` when you present to your cohort